PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : Chemical Equilibrium & Ionic Equilibrium (Question Bank # 01)

1. Ans. (D)

Sol. Let weak base is BOH

At equivalence point

eq. of acid = eq. of base

$$x \times \frac{2}{15} \times 1 = 2.5 \times \frac{2}{5} \times 1$$

x = 7.5 ml

$$BOH_{(aq)} + HCl_{(aq.)} \rightarrow BCl_{(aq)} + H_2O_{(\ell)}$$

2.5ml, $\frac{2}{5}$M 7.5 ml $\frac{2}{15}$M – –

1mm 1mm

– – 1mm

$$[BCl] = \frac{1}{10}M$$

[BCl] = salt of weak base and strong acid

$$B^+_{(aq)} + H_2O_{(\ell)} \rightleftharpoons BOH_{(aq.)} + H^+_{(aq)}$$

c(1–h) ch ch

$$K_h = \frac{10^{-14}}{10^{-12}} = \frac{ch^2}{1-h}$$

$$10^{-2} = 0.1 \times \frac{h^2}{1-h}$$

$h^2 + 0.1\,h - 0.1 = 0$

$$h = \frac{-0.1 + \sqrt{0.07 + 0.4}}{2}$$

$$h = \frac{-0.1 + 10^{-1} \times 6.4}{2} = 0.27$$

$[H^+] = ch = 0.1 \times 0.27 = 2.7 \times 10^{-2}$ M

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2. Ans. (D)

Sol. $Ag^{+}_{(aq)} + 2NH_{3(aq.)} \rightleftharpoons Ag(NH_3)_2^{+}{}_{(aq)}$

t = 0	0.8	x	–
t = t	5×10^{-8}	x –1.6	0.8

$$10^8 = \frac{0.8}{5\times10^{-8}\times(x-1.6)^2}$$

$(x - 1.6)^2 = 16 \times 10^{-2}$

$x - 1.6 = 0.4$

$x = 2M$

Moles of $NH_3 = (2 \times 2) = 4$.

3. Ans. (D)

Sol. Let salt be B^+A^- [$B^+A^-_{(aq)} \rightleftharpoons B^+_{(aq)} + A^-_{(aq)}$]

As salt is of strong base and weak acid hence anionic part undergoes hydrolysis.

$A^-_{(aq)} + H_2O_{(l)} \rightleftharpoons HA_{(aq)} + OH^-(aq)$ ..... (2)

In buffer of pH < 7, excess (H^+) ions combines with OH^- ions and makes reaction (2) to move forward and solubility (s_1) increases.

In buffer of pH > 7, excess (OH^-) in solution shifts reaction (2) in backward direction and hence solubility (s_2) decreases.

$\therefore\ s_2 > s_1 > s_3$

4. Ans. (C)

Sol. $LiCl_3.3NH_3(s) \rightleftharpoons LiCl.NH_3(s) + 2NH_3(g)$ $\quad K_p = 9\ atm^2$

$K_p = 9 = p^2_{NH_3} \Rightarrow p_{NH_3} = 3atm$ at equilibrium

moles of NH_3 at equilibrium

$3 \times 8.21 = n \times 0.0821 \times 300$

$n_{NH_3} = 1$ (at. eq.)

$\therefore$ moles of NH_3 converted = (2.8 – 1) = 1.8

moles of $LiCl_3.3NH_3$ formed = $\frac{1.8}{2}$ = 0.9

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5. Ans. (B)

Sol. $R_1OH + CH_3COOH \rightleftharpoons CH_3COOR_1 + H_2O \quad K_1$

(1–x) (1–x–y) x (x+y)

$R_2OH + CH_3COOH \rightleftharpoons CH_3COOR_2 + H_2O \quad K_2$

(1–y) (1–x–y) y (x+y)

$x + y = 0.8$ ..... (1)

$\frac{x}{y} = \frac{3}{2}$

$x = 0.48 \quad y = 0.52$

$k_1 = \frac{0.48 \times 0.8}{0.52 \times 0.2} = 3.69$

6. Ans. (A)

Sol. $A \rightleftharpoons B + C \quad 10^6 = K$

t = 0 1 – –

t = t – 1 1 (due to very high K)

$B + D \rightleftharpoons A \quad K = 10^{-6}$

t = 0 1 1 –

eq. 1–x 1–x x

$10^{-6} = \frac{x}{(1-x)(1-x)} \Rightarrow x = [A] = 10^{-6}\ M$

7. Ans. (C)

Sol. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq) \quad K_{sp} = 10^{-10}$

$Ag^+(aq) + 2Cl^-(aq) \rightleftharpoons [AgCl_2]^-(aq) \quad K_F = 3 \times 10^5$

$AgCl(s) + Cl^-(aq) \rightleftharpoons [AgCl_2]^-(aq) \quad 3 \times 10^{-5}$

0.01 mol a mol –

– (a – 0.01) mol 0.01 mol

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$$3 \times 10^{-5} = \frac{0.01/100}{(a-0.01)/100}$$

$$a - 0.01 = \frac{1000}{3}$$

$$a = \frac{1000}{3} \text{ moles} \Rightarrow \text{mass} = \frac{1000}{3} \times 58.5 = 19.5\ K_A$$

8. Ans. (B)

Sol. For salt of weak acid and weak base degree of hydrolysis does not depend on concentration.

9. Ans. (A, B, C)

Sol. For basic solution, if pH = 8 (assume)

(a) $pH > \frac{pK_w}{2} \Rightarrow pH > 7$

(b) pH > pOH

(c) $pOH < \frac{pK_w}{2} \Rightarrow pOH < 7$

10. Ans. (A,D)

Sol. (a) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ K_c

t = 0	1	1	–
eq.	(1–0.004)	(1+0.004)	0.004
	=1	=1	

$$K_C = \frac{\frac{0.004}{10} \times \frac{1}{10}}{\frac{1}{10}} = 0.0004\,M$$

(b) $K_P = 0.0004 \times (0.0821 \times 400)^1$

For backward reaction $K'_P = \frac{1}{K_P}$

(c) At equilibrium, $(\Delta G = 0)$ in both cases.

(d) Q_P remains same.

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11. Ans. (A,C)

Sol. As T increases $\Rightarrow$ K_w increases $\Rightarrow$ pH scale decreases.

pH at neutral point and pOH at neutral point decreases.

(a) As T increases $\Rightarrow$ K_w increases $\Rightarrow$ α increases

(b) At 25°C $\Rightarrow$ $K_w = 10^{-14}$ $\rightarrow$ pH of H_2O is 7.

At 90°C $\Rightarrow$ $K_w = 10^{-12}$ $\rightarrow$ pH of H_2O is 6.

At 60°C $\rightarrow$ pH scale is between 6 and 7.

(c) At 90°C, pH of water is 6, for acid pH < 7.

(d) Ionisation constant $K_i = \dfrac{K_w}{55.56}$

12. Ans. (A,B)

Sol. (a) As T increases $\Rightarrow$ K_w increases $\Rightarrow$ pH scale decreases

pH at neutral point decreases

pOH at neutral point decreases

Suppose at 25°C $K_w = 10^{-14}$

at 70°C $K_w = 10^{-13}$

at 70°C $\rightarrow$ pOH = 7

pH = (13 – 7) = 6 (acidic)

(b) Let (H^+) concentration changes from 10^{-3} M to 10^{-5} M

pH changes from 3 to 5

(c) Salt of SB + WA (CH_3COONa)

$[OH^-] = \sqrt{K_H \times C}$

As C decreases $\Rightarrow$ $[OH^-]$ decreases $\Rightarrow$ pOH increases $\Rightarrow$ pH decreases

(d) Buffer capacity = $\dfrac{\text{moles of } H^+ / OH^- \text{ required per litre of solution}}{\text{change in pH}}$

with dilution both numerator and denominator donot change proportionally.

13. Ans. (C, (D)

Sol. (a) (strong acid + weak acid) $\rightarrow$ donot form buffer

(c) Strong acid replaces weak acid from it's salt solution and if strong acid is L.R., we get buffer

$CH_3COONa + HNO_3 \rightarrow CH_3COOH + NaNO_3$

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SECTION-II: (Integer Second Decimal)

1. Ans. (3.00)

Sol. $A(g) \rightleftharpoons 2B(g)$

eq. 4 2 $K_P = \frac{2^2}{4} = 1$

At total pressure of 12 bar

Let pressure of 'B' at equilibrium is 'p' bar

then pressure of 'A' at equilibrium is (12 – p) bar

$$K_P = \frac{p^2}{12-p}$$

$$1 = \frac{p^2}{12-p}$$

$p^2 + p - 12 = 0$

$p = 3$ bar

2. Ans. (3.00)

Sol. $BOH(aq) + HCl(aq) \rightarrow BCl(aq) + H_2O(l)$

At $\frac{1}{4}$ eq. $\frac{3a}{4}$ $\frac{a}{4}$

Basic buffer $pOH = pK_b + \log \frac{[BCl]}{[BOH]}$

$$6 = pK_b + \log \frac{a/4}{3a/4}$$

$pK_b = (6 + \log 3)$

when enough strong base (like NaOH) is added

$BCl(aq) + NaOH(aq) \rightarrow BOH(aq) + NaCl(aq)$

a/4 3a/4

As per numerical $\frac{a}{4} = 7.5$, a = 30 millimoles

on complete conversion [BOH] = $\frac{30}{10}$ = 3M (weak base)

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$$pOH = \frac{1}{2}[pK_b - \log C]$$

$$= \frac{1}{2}[6 + \log 3 - \log 3] \Rightarrow pOH = 3$$

3. Ans. (2.94 to 3.00)

Sol. $pOH = 11$

$pH = 3$

$[H^+] = 10^{-3}$ M

$$[H^+] = \sqrt{K_a \times C}$$

$$10^{-3} = \sqrt{1.8\times10^{-5} \times \frac{10/60}{V}}$$

$V = 3L$

4. Ans. (16.10 to 16.30)

Sol. $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

(0.01)

$$B^+(aq) + H_2O(l) \rightleftharpoons BOH(aq) + H^+(aq)$$

$1-x$ $\quad$ x $\quad$ $(x + 0.01)$

$$\frac{x(x+0.01)}{1-x} = 10^{-4}$$

As $(1 - x) \simeq 1$

$x^2 + 10^{-2}\,x - 10^{-4} = 0$

$$x = \frac{-10^{-2} + \sqrt{10^{-4} + 4\times10^{-4}}}{2}$$

$$x = \frac{-10^{-2} + 2.23\times10^{-2}}{2} = 0.618 \times 10^{-2}$$

$(H^+) = (0.01 + x) = 1.618 \times 10^{-2}$

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SECTION–IV : Matrix-Match Type

1. Ans. (A)-P,Q,R,S ; (B) P, Q, R, S, (C)- P, Q, R, S,T, (D)-P,Q

Sol. (P) When $B_2(g)$ is added (product) all reaction equilibrium shift towards left hand side.

(Q) All reactions are dissociation reaction, hence endothermic, so on increasing temperature equilibrium shifts towards right and 'α' increases.

(R) On adding inert gas at constant pressure, volume of system increases hence concentration changes irrespective of direction of reaction.

For reaction $AB_2(s) \rightleftharpoons A_2(g) + B_2(g)$

concentration of A_2 and B_2 remains same as if concentration of A_2 and B_2 changes then K_p or K_c changes.

(S) P increases $\Rightarrow$ V decreases $\Rightarrow$ conc. of all increases except (D) option.

(T) As $\left(\Delta n_g = 0\right)$ for option (C).

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